On FilBench, the base checkpoints of Gemma 4 12B and Qwen3.5-9B are not separable overall: task-dependent strengths behind a pre-registered null

PinoAI research note · RESULT-002 · DEC-001 Part C

Stan Lee V. Natividad
Kooya PH AI Research
stan@kooya.ph

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Abstract

A companion note (RESULT-001) found that Gemma 4 12B's instruct version clearly outperforms Qwen3.5-9B and Sailor2-8B on a native Filipino and Taglish evaluation. The instruct versions include vendor post-training that an adapting project does not inherit, so we also compared the **base checkpoints** we would actually continue-pretrain. We used 12,956 pooled FilBench multiple-choice items per model, scored by log-likelihood, with a paired exact McNemar test and the same decision rule as before.

Registered result:

- Qwen3.5-9B and Gemma 4 12B each clearly beat Sailor2-8B ($p < 10^{-11}$).
- **Gemma 4 12B and Qwen3.5-9B are not significantly different** (1,081 vs 1,003 discordant wins, $p = 0.092$).
- So no model is favoured against both rivals.

Post-hoc analyses, labelled as such, explain the null:

- The pooled test is 56.7% one sentiment task, because we failed to cap it at registration.
- Excluding that task, the direction flips: Qwen narrowly leads, $p = 0.046$.
- Per task, Gemma leads on Cebuano reading comprehension, sentiment and culture. Qwen leads on named-entity and topic tasks.

At the base level the two models are task-dependent rather than ranked. Gemma's clear instruct-level advantage is therefore plausibly due in part to post-training.

1. Introduction

The base-model decision (DEC-001) registered three pieces of evidence and a reading order in advance:

- **Part A:** a native-authored evaluation of the instruct versions. Supported; see RESULT-001.
- **Part B:** tokenizer efficiency. Supported.
- **Part C:** a comparison of the base checkpoints themselves. This note.

Part C matters because an adaptation project inherits the base, not the vendor's instruction tuning. Under the registered reading order, Part C is consulted only if Part A selects no model, so this note does not change the decision. It does change how confidently the decision should be interpreted.

2. Registered method

Everything below was fixed before any base-model result existed:

- **Models:**
  - `sail/Sailor2-8B`
  - `Qwen/Qwen3.5-9B-Base`
  - `google/gemma-4-12B` Each is the base checkpoint, at a pinned revision.
- **Benchmark:** FilBench log-likelihood multiple choice [1].

- Tasks: the licence-cleared, non-gated, non-script-loader tasks, which gives 10 task families in Filipino, Tagalog and Cebuano.
- Settings: chat template off.
- Cap: Global-MMLU limited to the first 20 items per subject (1,140 items).
- Size: 12,956 items per model, with identical item sets across models, paired by task and document id.

- **Harness validation:** before the runs, the harness reproduced three published leaderboard scores of a reference model within about one standard error, on both platforms used.
- **Hardware:** one NVIDIA A100 80GB (PCIe) for all three models. Scoring is deterministic.
- **Test:** paired exact two-sided McNemar per pair, $\alpha = 0.05$. A model "is favoured" in a pair if it has more discordant wins and $p < 0.05$.

3. Results

3.1 Registered result

Table 1: Part C, registered analysis: 12,956 paired items per comparison. Source: `DEC-001-partC.txt`.

pair	n	wins A:B	both right	both wrong	McNemar p	rule favours
sailor2-8b vs qwen3.5-9b	12,956	661:939	7,928	3,428	< 0.0001	qwen3.5-9b
sailor2-8b vs gemma-4-12b	12,956	920:1,276	7,669	3,091	< 0.0001	gemma-4-12b
qwen3.5-9b vs gemma-4-12b	12,956	1,003:1,081	7,864	3,008	0.0916	no significant difference

No model is favoured in both of its pairs. Overall accuracy was 66.3% for Sailor2-8B (8,589), 68.4% for Qwen3.5-9B (8,867) and 69.0% for Gemma 4 12B (8,945).

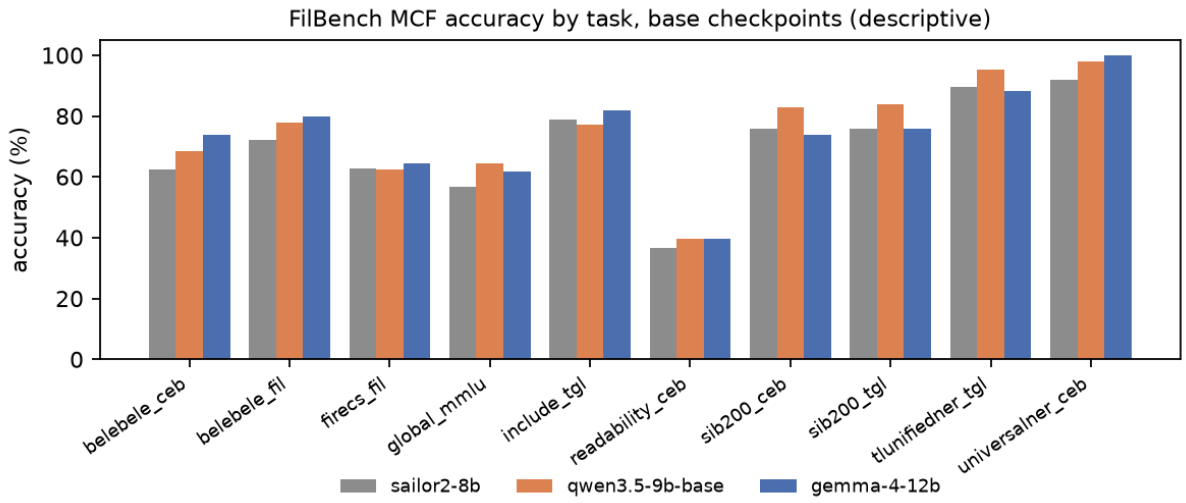


Figure 1: FilBench accuracy per task family, base checkpoints. Descriptive; the registered test pools all items.

Table 2: Accuracy (%) per task family, with each family's share of the pooled test. FiReCS (sentiment) is 56.7% of the pool.

task	n	share of pool	sailor2-8b	qwen3.5-9b-base	gemma-4-12b
belebele_ceb	900	6.9%	62.3	68.6	73.8
belebele_fil	900	6.9%	72.3	77.7	80.0
firecs_fil	7,340	56.7%	62.6	62.3	64.3
global_mmlu_all_tgl	1,140	8.8%	56.8	64.5	61.7
include_tgl	500	3.9%	78.8	77.0	81.8
readability_ceb	350	2.7%	36.6	39.7	39.7
sib200_ceb	99	0.8%	75.8	82.8	73.7
sib200_tgl	99	0.8%	75.8	83.8	75.8
tlunifiedner_tgl	1,579	12.2%	89.7	95.2	88.2
universalner_ceb	49	0.4%	91.8	98.0	100.0

3.2 Post-hoc analyses (descriptive; they cannot change 3.1)

Table 3: Post-hoc: pooled test excluding FiReCS (n = 5,616).

pair	n	wins A:B	McNemar p	favours
sailor2-8b vs qwen3.5-9b	5,616	487:786	< 0.0001	qwen3.5-9b
sailor2-8b vs gemma-4-12b	5,616	480:712	< 0.0001	gemma-4-12b
qwen3.5-9b vs gemma-4-12b	5,616	579:512	0.0456	qwen3.5-9b

Table 4: Post-hoc: Qwen3.5-9B vs Gemma 4 12B per task, uncorrected. With a Holm correction over the 10 tasks, three survive: TLUnified-NER (Qwen), FiReCS (Gemma) and Belebele-Cebuano (Gemma).

task	n	qwen only : gemma only	McNemar p (uncorrected)	favours
belebele_ceb	900	83:130	0.0016	gemma-4-12b
belebele_fil	900	72:93	0.1192	-
firecs_fil	7,340	424:569	< 0.0001	gemma-4-12b
global_mmlu_all_tgl	1,140	166:134	0.0733	-
include_tgl	500	30:54	0.0116	gemma-4-12b
readability_ceb	350	63:63	1.0000	-
sib200_ceb	99	13:4	0.0490	qwen3.5-9b
sib200_tgl	99	10:2	0.0386	qwen3.5-9b
tlunifiedner_tgl	1,579	142:31	< 0.0001	qwen3.5-9b
universalner_ceb	49	0:1	1.0000	-

4. Discussion

The base checkpoints of Gemma 4 12B and Qwen3.5-9B are task-dependent, not ranked.

- **The direction depends on the mix.** It flips depending on which large task dominates the pool:
  - Gemma leads, though not significantly, when the sentiment task is included.
  - Qwen leads narrowly when it is excluded.
- **Gemma's strengths:** reading comprehension (especially Cebuano), sentiment and cultural knowledge (INCLUDE).
- **Qwen's strengths:** token-level named-entity recognition and short topic classification.
- **Sailor2 trails both**, despite regional pretraining. That matches its weak instruct-level showing.

Two consequences for PinoAI.

1. **Part of Gemma's advantage may not transfer.** Gemma's large margin at the instruct level (RESULT-001) is plausibly due partly to its vendor's post-training, which a continued-pretraining project replaces. The registered decision still selects Gemma via Part A, but the expected advantage *after our own training* is less certain than Part A alone suggests.

2. **The strengths that matter most are Gemma's.** Gemma's base-level strengths, reading comprehension, Cebuano and cultural knowledge, are the ones most aligned with PinoAI's goals. The final choice still depends on compute feasibility (Gate F), which is being measured.

5. Limitations

- **Pool composition (our error).** The registration capped only Global-MMLU. FiReCS enters with its 7,340-item training split, as FilBench scores it, so it is 56.7% of the pooled items. We did not re-weight after seeing results. The post-hoc analyses show why this matters. Future pooled tests will cap every task at registration and state each task's share first.
- **Multiple-choice likelihood only.** It measures how a base model ranks given options, not generation.
- **Contamination.** Several FilBench sources are public and widely crawled (see our benchmark survey), so absolute accuracies may be inflated. Paired comparisons are less affected, but not immune.
- **Language coverage.** No Ilocano or Hiligaynon tasks are included.

6. Next steps

- Apply the compute-feasibility gate (Gate F) and close the base-model decision under the registered reading order.
- Test whether Gemma's reading and Cebuano strengths survive continued pretraining, as part of the planned ablations.

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References

[1] L. J. V. Miranda and others, "FilBench: Can LLMs Understand and Generate Filipino?," *arXiv preprint arXiv:2508.03523*, 2025.